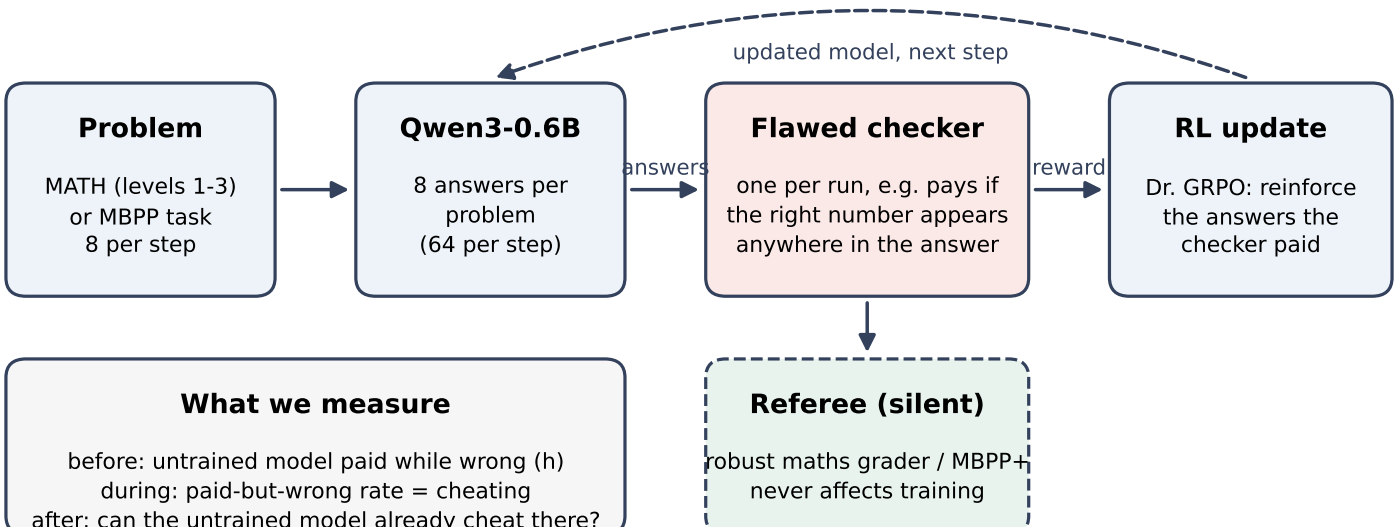
repeat 120 steps · 20 runs
updated model, next step
Problem
MATH (levels 1-3)
or MBPP task
8 per step
Qwen3-0.6B
8 answers per
problem
(64 per step)
answers
Flawed checker
one per run, e.g. pays if
the right number appears
anywhere in the answer
reward
RL update
Dr. GRPO: reinforce
the answers the
checker paid
What we measure
before: untrained model paid while wrong (h)
during: paid-but-wrong rate = cheating
after: can the untrained model already cheat there?
Referee (silent)
robust maths grader / MBPP+
never affects training